

Jingyu Huang

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Research Interests

LLM Agents, Reinforcement Learning for Agents, Multi-Agent Orchestration, Tool-Use & Browser-Based Agent Infrastructure, LLM Safety and Red-Teaming, Retrieval-Augmented Generation

Education

University of California, Los Angeles (UCLA) Fall 2026 (Admitted)
M.S. in Computer Science

University of Wisconsin–Madison Sept 2024 – May 2026
B.S. in Computer Science

- **GPA:** 3.9 / 4.0

- **Coursework:** Big Data Systems, Machine Learning, Introduction to AI

Beijing University of Posts and Telecommunications (BUPT) Sept 2021 – June 2024
B.S. in Telecommunications with Management

- **GPA:** 3.72 / 4.0

- **Coursework:** Data Structures, Machine Learning, Principles of Telecommunication

Research Experience

Research Intern Madison, WI
SafoLab, UW-Madison Aug 2024 – Mar 2026

- Designed **Multi-Turn-Jailbreaker**, a three-stage pipeline for automated multi-turn LLM jailbreaking: attack tree generation → supervised fine-tuning → GRPO-style RL optimization of the attacker policy. Paper submitted.
- Built a multi-GPU orchestration system managing rollout and training phases across GPU workers, with vLLM serving, DeepSpeed ZeRO training, and LlamaGuard-based reward evaluation.
- Developed configurable attack trees with variable branching depth and a strategy library; deployed attacker checkpoints to HuggingFace ([MartinJYHuang/JA-v1](#) 📄).
- Achieved 1st place (Round 2) at Amazon Nova AI Challenge; technique report published on Amazon Science.

Research Intern Beijing, China
Gamma Lab, BUPT Jan 2024 – Nov 2024

- Contributed to HGAT algorithm improvements in the GammaGL library for heterogeneous graph learning.

Work Experience

LLM Agent Engineer Shenzhen, China
Linksome – AI Recruitment Platform May 2026 – Sep 2026

- Designed and built a multi-stage agent pipeline for automated candidate screening: PDF parsing → parallel sub-agent research (school profiling, GitHub verification, publication validation) → synthesized screening report with recommendations.
- Conducted systematic experiments comparing LLM backends (DeepSeek, GPT-5.5, GPT-5.4) and agent modes to optimize analysis depth vs. token cost; identified critical failure modes in smaller models.
- Developed context-management strategies (cache-and-spill, session persistence, structured memory) and reusable skill definitions for standardized investigation workflows.

Projects

Amazon Nova AI Challenge

*Amazon Nova AI
Challenge Technique
Report* [🔗](#)

- Built multi-turn jailbreak agent with modular attacker strategies and planning; **1st place (Round 2)** and finalist presentation at Amazon Seattle HQ.
- Designed scalable red-teaming dataset pipeline for adversarial prompt generation and error-pattern analysis.

AgentBrowser – Browser Control Infrastructure for AI Agents

github.com/martinjingyu/AgentBrowser [🔗](#)

- Built a pip-installable Python library controlling Chrome via CDP over a Node.js bridge – navigate, click, type, scroll, and one-shot search helpers (Google/Bing/Baidu/Reddit) as first-class agent tools.
- Thread-safe architecture with per-thread isolated Chrome port, profile, and state; local profile reuse persists sessions across runs to reduce automated verification challenges.
- Ships OpenAI-compatible tool definitions for plug-and-play integration into any LLM tool-calling loop.

Agent-Tutorial – Self-Evolving Agent Framework with Guardian Architecture

github.com/martinjingyu/Agent-Tutorial [🔗](#)

- Built a **self-evolving agent framework** using a Guardian-Worker architecture: a thin parent process spawns the agent as a subprocess; when the agent modifies its own source code and calls `request_restart`, it exits with code 42 and the Guardian transparently re-spawns it with updated code.
- Implemented a **background self-review system** that runs as a non-blocking thread after each turn, analyzing the conversation transcript to propose memory or skill improvements without disrupting the main loop.
- Features include automatic context compaction with spill-to-disk, session save/resume, subprocess-based task delegation, a pluggable tool registry, and progressive tool blocking as the iteration limit approaches.

Technical Reports

- **Stepwise Multi-turn Jailbreak Attacks on Code LLMs via Task Decomposition and Test-time Scaling.**
Amazon Nova AI Challenge, 2025. [Link](#) [🔗](#)
- **Multi-Turn-Jailbreaker: Automated LLM Red-Teaming via Attack Tree Generation and GRPO Optimization.**
Ready-to-submit, 2026.

Technical Skills

- **Agent & RL:** Tool-use agent architecture, multi-agent orchestration, GRPO/RL fine-tuning, observe-plan-act loops, memory & self-reflection, prompt engineering, RAG
- **LLM Safety:** Multi-turn jailbreaking, attack tree generation, adversarial dataset construction, LlamaGuard evaluation
- **ML/DL Systems:** PyTorch, DeepSpeed, Transformers, vLLM, XGBoost, scikit-learn, Pandas
- **Languages & Tools:** Python, SQL, JavaScript, TypeScript, Bash, FastAPI, Playwright, CDP, Docker, multi-GPU training